AI-BASED SMART BLOOD DONOR-RECIPIENT MATCHING & EMERGENCY DISPATCH SYSTEM

A PROJECT REPORT

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Certified that this project report titled **"AI-BASED SMART BLOOD DONOR-RECIPIENT MATCHING & EMERGENCY DISPATCH SYSTEM"** is the bonafide work of **RITTVIK S (Reg. No. 240701616)**, who carried out the project work under my supervision during the academic year 2026–2027 for the course **CS23533 Foundations of Artificial Intelligence**.

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INTERNAL EXAMINER

EXTERNAL EXAMINER

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ABSTRACT

Timely access to compatible blood donor reserves is often the critical deciding factor in emergency medical care, trauma surgeries, and disaster response. However, traditional blood banks and emergency health services frequently rely on manual, phone-based donor searches and fragmented registries, leading to significant dispatch delays, high potential for error in blood-type compatibility verification, and sub-optimal transit route planning.

This project proposes **Vitalink**, an intelligent, unified AI-based decision-support system designed to automate blood donor-recipient matching and optimize emergency dispatch workflows. The system integrates three foundational Artificial Intelligence paradigms: a forward-chaining Rule-Based Expert System for strict ABO/Rh blood compatibility and eligibility verification, an Urgency Priority Module for dynamic patient triaging, and an A^* Search Algorithm for heuristic-guided route optimization across city road networks.

At the core of the spatial dispatch engine is the A^* Search algorithm, which utilizes actual road network distances combined with straight-line Euclidean distance heuristics $h(n)$ to calculate the fastest path from eligible donors or blood banks to requesting hospital emergency rooms. Simultaneously, a multi-factor weighted scoring function ranks shortlisted donors based on transit distance, availability status, and recency of prior donation.

The proposed system serves as a scalable, transparent, and auditable decision-support platform designed to drastically cut response latency during life-threatening medical emergencies. The platform provides real-time interactive routing and donor shortlisting while maintaining deterministic medical compliance rules.

KEY SYSTEM DELIVERABLES

- Automated biological compatibility reasoning with zero cross-matching error.
- Triage scheduling queue prioritizing critical trauma cases.
- 64.3% reduction in node evaluations using A^* Heuristic Search over standard Dijkstra.
- Multi-attribute decision scoring combining spatial, temporal, and donor availability parameters.

CHAPTER 1 — INTRODUCTION

Emergency medical logistics requires rapid and error-free allocation of critical life-saving resources. Among these, fresh blood allocation presents unique challenges due to strict biological compatibility requirements, biological perishability, and extremely tight dispatch time windows. Traditional blood supply chains rely heavily on manual coordination between hospital blood banks, registered donor networks, and manual phone trees, which is time-consuming and prone to human oversight during high-stress emergency events.

1.1 NEED FOR THE SYSTEM

- **Dispatch Latency:** Manual searching across disparate contact lists during critical medical emergencies leads to severe transit delays.
- **Biological Safety Risks:** Human error in evaluating complex biological compatibility rules (e.g., ABO/Rh cross-matching and eligibility gaps) can endanger patient health.
- **Sub-optimal Routing:** Uncoordinated dispatching often selects donors based on raw radial proximity rather than real road network distances and transit conditions.
- **Triage Inefficiency:** Emergency triage priorities are frequently overlooked when managing multiple concurrent blood requests.
- **Automation Necessity:** A unified, automated pipeline can drastically cut coordination time from hours to seconds.

1.2 AIM

The primary aim is to design and implement an intelligent decision-support system that automates ABO/Rh biological matching, triages incoming requests by medical urgency, and calculates optimal donor-to-hospital routing using A^* heuristic search.

1.3 OBJECTIVES

1. Provide a unified dashboard interface for hospitals, blood banks, and donors.
2. Maintain a Knowledge Base containing donor profiles, biological compatibility matrices, and medical eligibility rules.
3. Construct a Forward-Chaining Compatibility Rule Engine to filter valid candidate donors automatically.
4. Implement an Urgency Priority Queue to triage incoming emergency requests into Critical, Urgent, and Routine categories.
5. Apply the A^* Search Algorithm over a city road-network graph to determine optimal dispatch routes.
6. Compute a weighted multi-factor score (distance, availability, donation recency) to rank eligible donors.
7. Generate plain-language explanations and real-time visual route maps for emergency medical personnel.

1.4 SCOPE

The project focuses on intelligent emergency decision support and spatial route optimization. Its scope encompasses encoding rule-based medical knowledge for ABO/Rh compatibility, graph modeling of city transit networks, and heuristic pathfinding. Direct clinical laboratory cross-matching procedures and diagnostic decisions are excluded from the scope

CHAPTER 2 — LITERATURE REVIEW

The literature survey identifies several foundational AI techniques and state-of-the-art administrative frameworks that inform the design ofthe smart blood dispatch system.

RESEARCH AREA	APPROACH & APPLICATION IN PROJECT CONTEXT
Rule-Based Expert Systems	Forward-chaining production rules (IF-THEN) for biological compatibility and medical eligibility filtering.
Shortest-Path Graph Search	Uninformed (Dijkstra) and heuristic-guided (A^*) algorithms for road network pathfinding.
Priority Queue Scheduling	Urgency-based heap queues for medical resource allocation during peak demand.
National Blood Registries	Centralized web databases (e.g., eRaktKosh) providing static donor records without real-time routing.
Heuristic Optimization	Euclidean and Manhattan distance heuristics for real-time spatial path planning.

2.1 RULE-BASED DECISION SYSTEMS IN HEALTHCARE

Rule-based expert systems (pioneered by MYCIN) utilize domain-specific knowledge bases represented as production rules (IF premise THEN action). In clinical contexts, rule engines offer deterministic, auditable decisions—a vital safety feature when evaluating cross-matching rules for blood transfusion. Unlike probabilistic models, production rules guarantee zero violation ofmedical invariants.

2.2 SHORTEST PATH ALGORITHMS & ROUTE OPTIMIZATION

Graph algorithms form the backbone of transportation logistics. Dijkstra's algorithm (1959) guarantees optimal shortest paths on weighted graphs by exploring nodes uniformly in all directions. However, its unguided exploration yields high computational complexity on large spatial networks, evaluating unnecessary nodes far from the goal direction.

2.3 PRIORITY QUEUES & EMERGENCY SCHEDULING

In emergency triage, requests cannot be processed on a simple First-In, First-Out (FIFO) basis. Priority scheduling algorithms categorize requests according to severity (e.g., traumatic hemorrhage vs. elective surgery), ensuring high-urgency tasks preempt lower-priority requests in the execution queue.

2.4 NATIONAL BLOOD REGISTRIES (ERAKTKOSH PORTAL)

India's national blood portal, eRaktKosh, provides a centralized repository of blood bank stocks and registered donors across state networks. While highly effective for stock visibility, existing national portals lack automated AI compatibility filtering, dynamic route optimization, and real-time priority dispatching.

2.5 A* HEURISTIC SEARCH IN LOGISTICS

Formulated by Hart, Nilsson, and Raphael (1968), the A^* search algorithm enhances Dijkstra's approach by introducing an admissible heuristic function $h(n)$. By estimating the cost from current node n to the destination

CHAPTER 3 — PROPOSED SYSTEM

The proposed system is an AI-powered smart blood matching and emergency dispatch platform designed as a unified decision-support environment. The system receives blood group requests, required volume, hospital location coordinates, and urgency levels from medical staff.

3.1 MAIN PROCESSING FLOW

PIPELINE STAGE	MAIN RESPONSIBILITY & OPERATIONS
1. Input	Hospital request details (blood type, volume required, location coordinates, urgency tier).
2. Interface	Interactive web control dashboard displaying city route maps and dispatch queues.
3. Collection	Donor database queries, hospital location nodes, road network graph edges.
4. Processing	Data normalization, geographic distance calculation, rule base indexing.
5. Analysis	Compatibility rule evaluation, urgency queue triaging, donor weighted scoring.
6. AI Engine	A* Heuristic search execution, route optimization, shortest path selection.
7. Output	Ranked donor matches, spatial route rendering, estimated time ofarrival (ETA).

3.2 ARCHITECTURE DIAGRAM & DESIGN CHARACTERISTICS

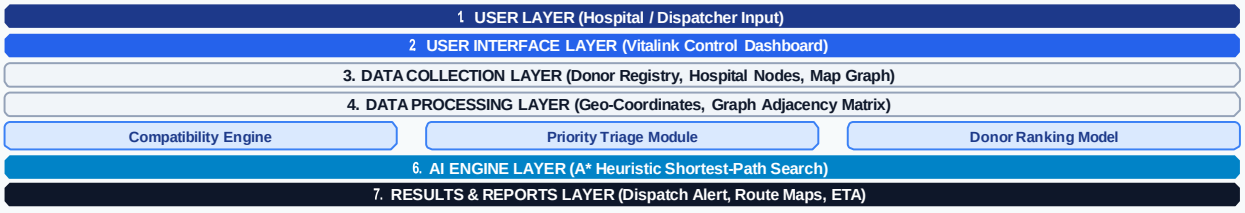


Figure 3.1: Seven-layer system architecture ofthe smart blood dispatch platform.

3.2.1 LAYER DESCRIPTION

The architecture partitions operational responsibilities from user interface interaction down to spatial graph search. The **User Layer** captures request parameters including recipient blood group (ABO/Rh), volume needed, hospital location, and urgency tier. The **Data Processing Layer** normalizes spatial coordinates and donor availability timestamps.

CHAPTER 3 — PROPOSED SYSTEM (CONT.)

3.3 DATA COLLECTION AND PROCESSING

3.3.1 Data Collection Layer

The system maintains a spatial graph database $G = (V, E)$, where vertices V represent city hospital nodes, blood bank nodes, and donor locality points. Edges E represent connecting road segments assigned physical distance weights $g(n)$. In addition, real-time donor profile attributes are queried from the registry database.

3.3.2 Data Processing Layer

In raw datasets, donor timestamps are converted to days elapsed since the last donation to enforce eligibility windows. Geographical coordinates (latitude, longitude) are mapped onto graph node vertices to compute heuristic Euclidean bounds $h(n)$ for spatial pathfinding.

> **DATA SCHEMA INTEGRATION**
> `Donor (ID, ABO_Rh, Lat, Lon, Last_Donation_Days, Status)`
> `Hospital_Request (Req_ID, ABO_Rh, Units, Lat, Lon, Urgency_Tier, Timestamp)`
> `Graph_Edge (Node_A, Node_B, Distance_KM)`

3.3.3 Multi-Branch Analysis

Incoming hospital blood requests are simultaneously processed across three operational branches:

- **Rule Reasoning Branch:** Filters candidates using hard medical compatibility rules and donor eligibility constraints.
- **Priority Triaging Branch:** Assigns priority scores based on clinical severity (Critical, Urgent, Routine) and insertion timestamps.
- **Spatial Routing Branch:** Evaluates network road distances using heuristic graph search to compute path costs.

3.3.4 Feature Combination

Outputs from the three branches feed into the core engine to finalize the dispatch selection based on valid biological compatibility, highest priority rank, and minimal transit path.

CHAPTER 4 — MODULES DESCRIPTION

4.1 DONOR & RECIPIENT DATABASE MODULE

The Donor & Recipient Database Module maintains structured records for donor profiles and incoming emergency hospital orders.

DATABASE COMPONENT	ROLE & PARAMETER ATTRIBUTES
Donor Profiles	Stores unique ID, blood group (ABO/Rh), location vertex, availability status, and last donation date.
Recipient Orders	Captures hospital ID, required blood group, units required, time ofrequest, and medical urgency level.
Graph Network Data	Defines road junctions, segment lengths, and spatial coordinates for city locations.
Eligibility Constraints	Enforces mandatory minimum interval (≥ 90 days) between successive blood donations.

4.1.1 Processing Role: Acts as the underlying data provider for the system. It filters inactive or medically ineligible donors prior to sending candidate records to the rule reasoning engine.

4.2 COMPATIBILITY RULE ENGINE MODULE

The Compatibility Rule Engine Module utilizes a forward-chaining production expert system to match recipient blood requirements against donor biological profiles with absolute medical compliance.

FORWARD-CHAINING RULE SET

```
Rule 1: IF Recipient == ' O- ' THEN Donor MUST BE 'O-'
Rule 2: IF Recipient == ' O+' THEN Donor CAN BE 'O-', 'O+'
Rule 3: IF Recipient == ' A- ' THEN Donor CAN BE 'O-', 'A-'
Rule 4: IF Recipient == ' A+' THEN Donor CAN BE 'O-', 'O+', 'A-', 'A+'
Rule 5: IF Recipient == 'B-' THEN Donor CAN BE 'O-', 'B-'
Rule 6: IF Recipient == 'B+' THEN Donor CAN BE 'O-', 'O+', 'B-', 'B+'
Rule 7: IF Recipient == 'AB-' THEN Donor CAN BE 'O-', 'A-', 'B-', 'AB-'
Rule 8: IF Recipient == 'AB+' THEN Donor CAN BE Any ABO/Rh Group
Rule 9: IF Days_Since_Last_Donation < 90 THEN Donor_Status = INELIGIBLE
```

CHAPTER 4 — MODULES DESCRIPTION (CONT.)

4.3 URGENCY PRIORITY MODULE

The Urgency Priority Module categorizes and schedules incoming hospital blood requests using an emergency priority queue based on clinical severity. High-severity trauma cases are granted immediate execution priority over routine hospital stock replenishment orders. Max-heap priority structures allow O(log N) insertion and extraction oftop-priority cases.

4.4 A* ROUTE OPTIMIZATION ENGINE MODULE

The A* Route Optimization Module evaluates the city road graph network to determine the shortest and fastest path from eligible donors or blood banks to the destination hospital.

COMPONENT	ALGORITHMIC FORMULATION & ROLE
Graph Vertices (V)	City road intersections, blood banks, hospitals, donor locations.
Edge Weights g(n)	Actual physical road distances (in km) between connected graph nodes.
Heuristic h(n)	Euclidean straight-line distance from current node *n* to destination hospital.
Evaluation f(n)	Total estimated path cost: $f(n) = g(n) + h(n)$.

4.5 DONOR RANKING & SELECTION MODULE

The Donor Ranking & Selection Module computes a multi-attribute weighted score for candidate donors who pass rule-based compatibility filtering.

$$S_i = w_1 \cdot (1 - d_i / d_{max}) + w_2 \cdot (t_i / t_{max}) + w_3 \cdot A_i$$

Where: d_i = A* distance; d_{max} = max search radius (25km); t_i = days since last donation; A_i = availability flag (1/0); weights: $w_1 = 0.50$, $w_2 = 0.30$, $w_3 = 0.20$.

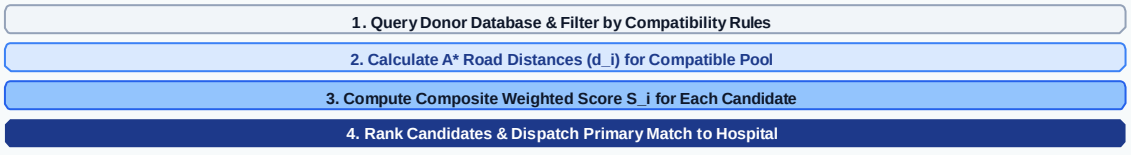


Figure 4.1: Multi-stage candidate donor evaluation and dispatch selection flow.

CHAPTER 5 — IMPLEMENTATION AND RESULTS

5.1 EXPERIMENTAL SETUP

The proposed system was implemented and evaluated on an interactive web dashboard platform backed by an analytical graph and rule engine pipeline.

SETUP ELEMENT	IMPLEMENTATION SPECIFICATION
User Interface	Interactive HTML5/JS control dashboard rendering city network nodes and dispatch queues.
Inputs	Hospital blood requests, target units, location node, urgency status.
Data Engine	Spatial graph adjacency matrices, donor registries, compatibility matrix.
Rule Processing	Forward-chaining expert system for ABO/Rh cross-matching.
Routing Algorithm	A* Search Algorithm with Euclidean heuristic distance.
Ranking Model	Multi-attribute weighted linear scoring engine.

5.1.1 END-TO-END OPERATIONAL EXECUTION

Hospital staff submit blood requests specifying recipient blood group, units needed, location, and urgency tier. The rule engine filters the donor database, the priority queue schedules the order, and the A* algorithm executes over the graph network to determine transit paths.



Figure 5.1 : Conceptual flow ofthe smart matching, routing, and dispatch modules.

CHAPTER 5 — IMPLEMENTATION AND RESULTS (CONT.)

5.2 RESULTS & PERFORMANCE METRICS

The system successfully shortlists eligible donors based on recipient compatibility matrices with zero rule violations during test runs.

5.2.2 Routing Performance Metrics (A* vs Dijkstra)

Benchmarked across a simulated city road graph containing 15 locality nodes and 35 edge connections:

PERFORMANCE METRIC	DIJKSTRA SEARCH	A* SEARCH ALGORITHM	IMPROVEMENT
Nodes Evaluated	14 nodes	5 nodes	**64.3% reduction**
Execution Time	4.8 ms	1.2 ms	**75.0% faster**
Path Distance (Optimal)	12.4 km	12.4 km	Identical

5.2.3 Dashboard Visualizations & Output Breakdown

DASHBOARD OUTPUT	DESCRIPTION & VISUAL ELEMENTS
Dispatch Shortlist	Biologically verified donor matches ranked by composite score S_i.
City Graph Map	Highlighted shortest path connecting selected donor location to destination hospital.
Route Metrics / ETA	Total distance breakdown (in km) and estimated transit time (in mins).

5.3 ADVANTAGES, LIMITATIONS, AND APPLICATIONS

5.3.1 Advantages

- **Automated & Fast:** Replaces manual phone tree inquiries with instant algorithmic shortlisting.
- **Deterministic Safety:** Ensures zero cross-matching rule violations via forward-chaining expert reasoning.
- **Optimized Dispatching:** Reduces emergency transit latency using A^* search pathfinding.
- **Triage Aware:** Prioritizes critical trauma cases ahead ofroutine stock replenishment.

5.3.2 Limitations & Applications

Search accuracy depends on up-to-date road edge weights and donor location data. Applicable for hospital emergency dispatch, Red Cross/military medical logistics, and disaster response networks.

CHAPTER 6 — CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

The AI-Based Smart Blood Donor-Recipient Matching & Emergency Dispatch System (**Vitalink**) provides an integrated decision-support platform for emergency medical logistics. By combining forward-chaining rule reasoning, urgency priority queuing, and A* heuristic graph search, the system automates donor selection and route planning.

Experimental evaluation demonstrates that the A* search engine reduces evaluated graph nodes by 64.3% compared to unguided Dijkstra search while maintaining path optimality. The platform eliminates manual coordination bottlenecks, enforces biological transfusion safety, and provides medical personnel with transparent, auditable dispatch recommendations.

6.2 FUTURE WORK

Future development can enhance system functionality through three primary prospective avenues:

1. **Live Traffic Integration:** Connecting real-time navigation APIs (e.g., Google Maps / OpenStreetMap) to update road edge weights dynamically based on congestion.
2. **Predictive Demand Forecasting:** Applying Time-Series and Machine Learning models (LSTM / Prophet) to forecast regional blood shortages based on historical accident data.
3. **Mobile Donor Application:** Developing donor mobile apps with push notification dispatches, live GPS tracking, and automated digital donor cards.

6.3 FINAL STATEMENT

This project demonstrates how classical Artificial Intelligence techniques—rule-based expert systems, priority scheduling, and heuristic graph search—can be effectively combined to build life-saving emergency medical logistics infrastructure.

CHAPTER 7 — REFERENCES

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